Muvvala Sairam Suhas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.16
Narayana JR College Santhoshnagar, Mehdipatnam	Class XII (Telangana State Board)	2023	97.50%
Atomic Energy Central School Kaiga	Class X (CBSE)	2021	97.60%

WORK EXPERIENCE

Oracle Software Engineering Intern

- Contributed to **ACATS (Automated Customer Analysis & Test Solutioning)**, an AI-driven workflow analyzing customer bug evidence to identify deployed features, weak areas and targeted testing opportunities.
- Engineered an **agentic generator-validator pipeline** with Python orchestration and iterative repair, selectively expanding or repairing source-backed feature intelligence using validator feedback.
- Achieved **~89.7% lower input-token usage** in narrow repairs and **~80.5% lower average usage** in later generator iterations through workflow and context optimization.

PROJECTS

Parallelising Closeness Centrality | Research Project under Prof. Sathya Peri | C++

- Developed a scalable approach for **dynamic closeness centrality** using BCC decomposition to restrict recomputation to nodes affected by graph updates.
- Implemented **64-bit bit-parallel SpMM BFS** with OpenMP and Block-Cut Tree distance assembly for efficient parallel graph analytics.

Distributed Bitcoin Mining System | Course Project Under Prof. Sathya Peri| C++

- Built an **MPI-based distributed Bitcoin miner** using SHA-256, with Raft-based leader election, dynamic nonce-range allocation, worker checkpointing and fault-tolerant work reassignment across processes and machines.
- Implemented and **experimentally compared static partitioning, dynamic work-stealing and MPI-based mining**, analyzing performance across thread/process counts, mining difficulty and multi-VM deployments.

AFT -- Another Frequency Thing | Course Project under Prof. Ramakrishna Upadhastra | C++, Flex, Bison, LLVM

- Built a **statically-typed DSL compiler for frequency and signal-processing applications**, implementing the pipeline from lexical analysis and parsing through AST construction, semantic analysis and LLVM IR generation.
- Implemented **symbol tables, scope management and type checking** for numerical and signal-oriented data types including vectors, complex numbers and floats, with functions, recursion and control-flow constructs.

Intelligence-Driven Observability Platform | Course Project under Prof. Manish Singh

- Designed and implemented a **scalable end-to-end observability platform** for real-time logs, metrics and traces using Envoy, Vector, Redpanda, Flink and ClickHouse with hot/cold telemetry pipelines.
- Integrated **ML-powered anomaly detection, tail sampling and intelligent routing** with a Next.js monitoring dashboard, supporting real-time alerts, metrics, traces and multi-tenant telemetry analysis.

RISC-V Simulator with Cache Simulation | Course Project under Prof. Rajesh Kedia | C++

- Developed a **C++ RISC-V simulator** supporting step-wise execution, register/memory inspection, breakpoints, and configurable cache simulation across different sizes, associativities and replacement policies.
- Implemented **ISA instruction parsing, cache mapping and replacement logic** with a custom lexer/parser and interactive query-response interface; selected for peer and faculty presentation.

SKILLS

- **Languages:** C++, C, Python, Assembly (RISC-V), Solidity, SQL
- **Python Libraries:** NumPy, Pandas, Scikit-learn, PyTorch
- **Core CS:** Data Structures & Algorithms, OOP, Distributed Systems, Computer Architecture, Operating Systems
- **Systems & Tools:** Linux, MPI, Git, Docker, Bash, LLVM/Clang
- **Technologies:** Django, Flex, Bison, PostgreSQL, MongoDB

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Discrete Mathematics, Data Structures, Algorithms, Operating Systems I & II, Database Management Systems I & II, Computer Networks, Distributed Computing, Computer Architecture, Theory of Computation, Compilers I & II, Software Engineering, Parallel & Concurrent Programming, Quantum Computing, Fraud Analytics
- **Mathematics & AI:** Probability and Random Variables, Linear Algebra, Introduction to Statistics, Convex Optimization, Foundations of Machine Learning, Deep Learning

SCHOLASTIC ACHIEVEMENTS

- Top 15 in OPTIVER Tradethon(2025) at IIT Hyderabad
- Achieved Karnataka State Rank **20** in **NTSE** (Stage 1 2021) and advanced to the National Level.

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant - Introduction to Programming** under Prof. Jyothi Vedurada
- **Teaching Assistant - Discrete Mathematics** under Prof. Rogers Mathew
- **OCS Coordinator** - Pitching Coordinator (2024-25); Internship Coordinator (2025-26)
- **Blockspace Club - Core** - Explored blockchain fundamentals and smart contracts using Solidity

EXTRACURRICULAR

Served as the NCC (National Cadet Corps) leader at my school and successfully completed the 'A' Certificate training.

I enjoy playing Squash and Table Tennis in my free time.